

HW5, due May 7, 2026

1. Suppose there are six people—John, Mary, Rose, Steve, Ted, and Wendy—who pass rumors among themselves. Each day Jon talks with Mary and Wendy; Mary talks with John, Rose, and Steve; Rose talks with Mary, Steve, and Ted; Steve talks with Mary, Rose, Ted, and Wendy; Ted talks with Rose, Steve, and Wendy; and Wendy talks with John, Steve, and Ted. Whatever people hear one day they pass on to the others the next day.

(a) Model this rumor-passing situation with a graph. (Draw the graph and label the vertices.)

(b) How many days does it take to pass a rumor from John to Steve?

(c) Is there any way that if two people stopped talking to each other, then it makes take three days to pass a rumor from one person to all the others?

2.

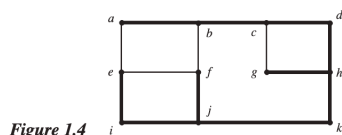


Figure 1.4

(a) An adjacency matrix for a graph is a matrix with n vertices is an $n \times n$ matrix where row i , column j is 1 if nodes i and j are adjacent in a graph; otherwise the entry for row i , column j is 0. Give the adjacency matrix for the graph in Figure 1.4.

(b) The distance matrix associated with a graph with n vertices is an $n \times n$ matrix where the (i, j) th entry (row i , column j) is $d(i, j)$, where $d(i, j)$ is the distance between two nodes. Here, the distance between two nodes is the minimum number of edges on a path from vertex i to vertex j . For example, for Figure 1.4, $d(a, k) = 4$. Write the distance matrix for the graph in Figure 1.4.

3. Suppose eight current varieties of chipmunk (names 1–8) evolved from a common ancestral strain through an evolutionary process in which at various stages one ancestral variety split into two varieties (none of the ancestral varieties survive when they split into two new varieties). The resulting tree is binary (root has out-degree 1, in-degree 0; leaves have out-degree 0, in-degree 1; and remaining nodes have in-degree 1 and out-degree 2).

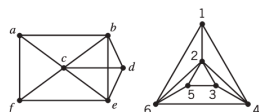
(a) Give an example of a tree that minimizes the maximum distance from each leaf to the root. (If this is confusing, try some trees of different shapes, and write the maximum distance from a

leaf to the root—figure out which tree minimizes these distances).

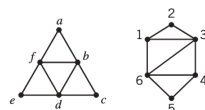
(b) What is the maximum possible distance from a leaf to the root? Give an example of a tree that achieves this maximum.

4. Which of the following pairs of graphs are isomorphic? Explain carefully and provide an isomorphism if the graphs are isomorphic or a reason that they cannot be isomorphic.

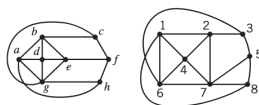
(a)



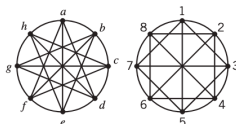
(b)



(c)



(d)



5. Are the following directed graphs isomorphic? Provide an isomorphism if they are or an argument for why they are not isomorphic. Give an answer for each of the $\binom{4}{2} = 6$ pairs of graphs.

